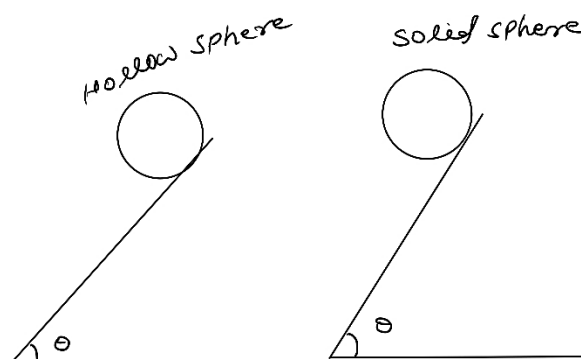


PHYSICS

- Q1** Hollow sphere of mass M and radius R reaches at bottom of the inclined plane in time t_1 and solid sphere of mass M and radius R reaches at bottom of the inclined plane in time t_2 . Find t_1/t_2 .



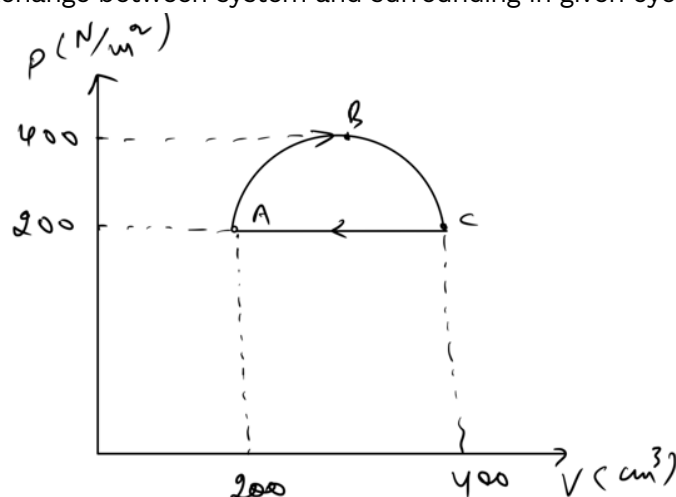
Sol.

$$t = \sqrt{\frac{2\ell \left(1 + \frac{K^2}{R^2}\right)}{g \sin \theta}}$$

$$\frac{t_1}{t_2} = \sqrt{\frac{\left(1 + \frac{K_1^2}{R_1^2}\right)}{\left(1 + \frac{K_2^2}{R_2^2}\right)}}$$

$$\frac{t_1}{t_2} = \sqrt{\frac{\left(1 + \frac{2}{3}\right)}{\left(1 + \frac{2}{5}\right)}} = \sqrt{\frac{25}{21}}$$

- Q2** Find out the heat exchange between system and surrounding in given cyclic process.



Sol.

$$\Delta U = 0$$

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta W = \text{Area under curve}$$

$$W = \frac{\pi R^2}{2}$$

$$= \frac{\pi}{2} \times 100 \times 10^{-6} \times 100$$

$$= \frac{\pi}{2} \times 10^{-2}$$

$$= \frac{\pi}{200} \text{ J}$$

Q3 A conical pendulum of mass 'm' and length ' ℓ ' moving with constant speed ' $\pi/3$ ' m/sec. Find out the tension in the string.

Sol. $T \sin \theta = \frac{mv^2}{r}$... (i)

$$T \cos \theta = mg \quad \dots (ii)$$

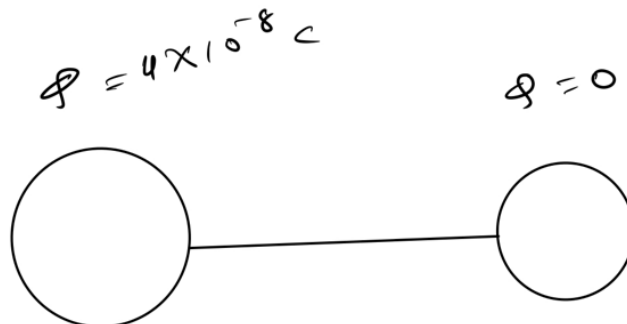
$$\sin \theta = \frac{r}{\ell}$$

Square and add equation (i) and (ii)

$$T = \sqrt{\left(\frac{mv^2}{r}\right)^2 + (mg)^2}$$

$$= \sqrt{\left(\frac{m\pi^2}{9\ell \sin \theta}\right)^2 + m^2 g^2}$$

Q4 One sphere is charged with $Q = 4 \times 10^{-8} \text{ C}$ and other is initially uncharged. After connecting them with wires they experience a force $9 \times 10^{-3} \text{ N}$. Find distance between them. (Both the spheres are identical)



Sol. $V_1 = V_2$

$$\frac{kx}{r_0} = \frac{k(Q-x)}{r_0}$$

$$x = \frac{Q}{2}$$

$$F = \frac{kQ_1Q_2}{r^2} = \frac{9 \times 10^9 \times Q^2}{4r^2}$$

$$r = 2 \text{ cm}$$

Q5 A body having bulk modulus $B = 2 \times 10^9 \text{ N/m}^2$ is subjected to a pressure P and undergoes change in volume of 2 percent. The change in pressure is:

Sol. $B = 2 \times 10^9 \text{ N/m}^2$

$$\% \text{ age change in vol.} = 2\% \quad \frac{\Delta V}{V} = \frac{2}{100} = 0.02$$

$$B = \frac{\Delta P}{\frac{\Delta V}{V}}$$

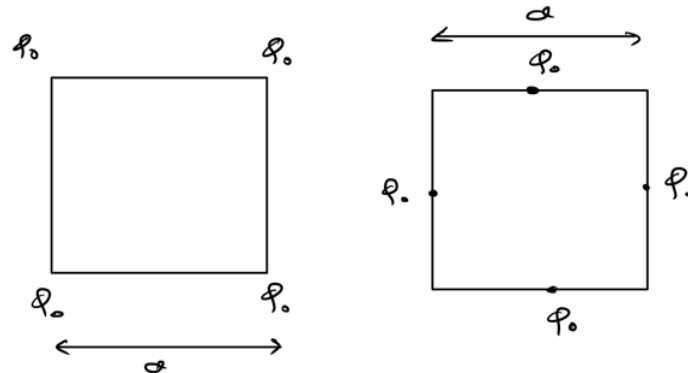
$$\Delta P = B \times \frac{\Delta V}{V}$$

$$\Delta P = (2 \times 10^9) \times (0.02)$$

$$\Delta P = 4 \times 10^7 \text{ N/m}^2$$

$$\Delta P = 40 \text{ MPa}$$

Q6 The potential difference at the center of the square in two setups.



Sol.

$$U_i = \frac{kQ_0^2}{a} + \frac{kQ_0^2}{a} + \frac{kQ_0^2}{\sqrt{2}a} + 2 \left(\frac{kQ_0^2}{a} \right) + \frac{kQ_0^2}{\sqrt{2}a}$$

$$= \frac{4kQ_0^2}{a} + \frac{\sqrt{2}kQ_0^2}{a} = \frac{kQ_0^2}{a} (4 + \sqrt{2})$$

$$U_f = \frac{kQ_0^2 \sqrt{2}}{a} (4 + \sqrt{2})$$

$$\Delta U = U_f - U_i = \frac{kQ_0^2}{a} (4 + \sqrt{2}) (0.41)$$

Q7 A solid sphere is performing pure rolling motion on fixed horizontal surface. Find ratio of $\frac{\text{Translational K.E.}}{\text{Rotational K.E.}}$.

Sol. For Pure Rolling : velocity COM & (ω) satisfy:

$$v = \omega R$$

$$KE_{\text{translational}} = \frac{1}{2} M v^2$$

$$\text{M.I. for solid sphere } I = \frac{2}{5} M R^2$$

$$KE_{\text{Rotational}} = \frac{1}{2} I \omega^2$$

Substitute $I = \frac{2}{5} M R^2$ & using rolling condition $v = \omega R$

$$KE_{\text{Rotation}} = \frac{1}{2} \left(\frac{2}{5} M R^2 \right) \left(\frac{v}{R} \right)^2$$

$$KE_{\text{Rotation}} = \frac{1}{2} \times \frac{2}{5} M v^2$$

$$KE_{\text{Rotation}} = \frac{1}{5} Mv^2$$

$$\text{Ratio : } \frac{KE_{\text{Rot}}}{KE_{\text{tan s}}} = \frac{1/2 Mv^2}{1/5 Mv^2}$$

$$\frac{KE_{\text{Rot}}}{KE_{\text{tan s}}} = \frac{1/2}{1/5}$$

Q8 Two monochromatic sources of light emitting wavelength 600 nm and 300 nm respectively. The ratio of the power of the sources $P_1/P_2 = 2$. If the rate of photon emission for the first source is $6 \times 10^9 \text{sec}^{-1}$, then the rate of photon emission for the second source is:

Sol. $\lambda_1 = 600 \text{ nm} = 600 \times 10^{-9} \text{ m}$
 $\lambda_2 = 300 \text{ nm} = 300 \times 10^{-9} \text{ m}$

$$\text{Power Ratio : } \frac{P_1}{P_2} = 2$$

$$n_1 = 6 \times 10^9 \text{ Photons/sec}$$

$$E = \frac{hc}{\lambda}$$

$$E_1 = \frac{hc}{\lambda_1}$$

$$E_2 = \frac{hc}{\lambda_2}$$

$$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{300}{600} = \frac{1}{2}$$

$$\therefore E_2 = 2E_1$$

$$P_1 = n_1 E_1$$

$$P_2 = n_2 E_2$$

Dividing power

$$\frac{P_1}{P_2} = \frac{n_1 E_1}{n_2 E_2}$$

Substituting the Ratio:

$$\frac{P_1}{P_2} = 2$$

$$2 = \frac{n_1}{n_2} \cdot \frac{E_1}{2E_1}$$

$$2 = \frac{n_1}{2n_2}$$

$$4n_2 = n_1$$

$$n_2 = \frac{n_1}{4}$$

$$n_2 = \frac{6 \times 10^9}{4}$$

$$n_2 = 1.5 \times 10^9 \text{ Photon/sec}$$

Q9 A farm field of 400 km^2 is viewed from a drone camera and the image of the field is shown on the screen of drone. The size of image is $2 \text{ cm} \times 2 \text{ cm}$. Find the focal length of the lens of the drone. Given drone is situated at a height of 12 km from ground.

Sol. $|m| = \frac{2 \times 10^{-2}}{20 \times 10^3}$

$$|m| = 10^{-6}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{u + f}{fu}$$

$$\frac{v}{u} = \left| \frac{f}{f + u} \right| m$$

$$\frac{f}{f + u} = -10^6$$

$$\frac{f}{f - 12 \times 10^3} = -10^6$$

$$f = -10^6 f + 12 \times 10^{-3}$$

$$f(1 + 10^{-6}) = 12 \times 10^{-3}$$

$$f = 12 \times 10^{-3}$$

$$f = 12 \text{ cm}$$

Q10 If temperature of body is decreased from 40° C to 28° C in 4 minutes. Then the temperature of the body after 4 more minutes is (Take temperature of surroundings = 16° C)

Sol. $\log \left(\frac{T_f - T_0}{T_i - T_0} \right) = K T$

$$\log \left(\frac{28 - 16}{40 - 16} \right) = K 4$$

$$\log \left(\frac{T - 16}{28 - 16} \right) = K 4$$

$$\frac{\log 12 - \log}{\frac{28 - 16}{40 - 16}} = \frac{T - 16}{28 - 16}$$

$$\frac{12}{24} = \frac{T - 16}{12}$$

$$T = 12$$

Q11 A particle is performing SHM with equation $x = x_0 \sin^2 \left(\frac{t}{2} \right)$, Where $x_0 = 1 \text{ m}$. Find the kinetic energy of particle as a function of time.

Sol. $x = x_0 \sin^2 \left(\frac{t}{2} \right)$

$$\frac{dx}{dt} = v = x_0 2 \sin \left(\frac{t}{2} \right) \cos \left(\frac{t}{2} \right)$$

$$= x_0 \sin t$$

$$v = \sin t$$

$$KE = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2}m \sin^2 t$$

Q12 If the acceleration due to gravity on the surface of earth is g , then acceleration due to gravity on a planet whose diameter is $1/3$ of that of earth and same mass as that of earth is $g' = ng$, where n is:

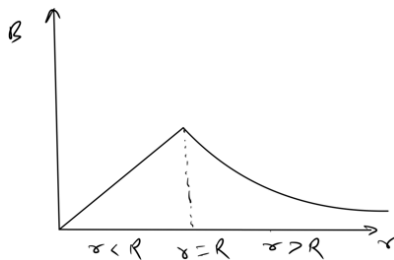
Sol. $g = \frac{GM_e}{R^2}$

$$g' = \frac{GM_e 9}{R^2}$$

$$g' = 9g$$

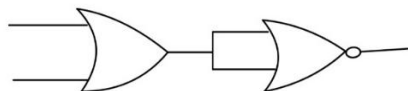
$$x = 9$$

Q13 An infinite wire of radius a carries a steady current. The graph of magnetic field v/s distance from the axis of wire.



Sol.

Q14 In which situation output will be zero.



(1) 0, 0

(2) 1, 0

(3) 0, 1

(4) 1, 1

Sol.

A	B	C	C	C + C	C + C
0	0	0	0	0	1
0	1	1	1	1	0
1	0	1	1	1	0
1	0	1	1	1	0

Q15 Find the angle made by velocity vector with y axis at $t = 2\text{sec}$. $\vec{r} = 3t^2\hat{i} - 5t\hat{j}$

Sol. $\vec{r} = 3t^2\hat{i} - 5t\hat{j}$

$$\vec{v} = 6t\hat{i} - 5\hat{j}$$

$$\vec{v} = 12\hat{i} - 5\hat{j}$$

Q16 If E , p , m and c denote the energy, linear momentum, mass and speed of light, then the equation representing the correct relation will be:

- (1) $E^2 = p^2c^2 + m^2c^4$
- (2) $E^2 = pc^2 + m^2c^4$
- (3) $E = p^2c^2 + m^2c^2$
- (4) $E^2 = pc^2 + m^2c^2$

Sol. $E = \sqrt{(m_0c^2)^2 + (pc)^2}$
 $E^2 = m_0^2c^4 + p^2c^2$

Q17 Assertion: Path of electron is straight line in uniform magnetic field.

Reason: $\vec{v}$ and $\vec{B}$ are in same direction.

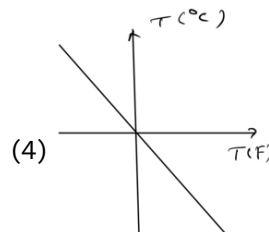
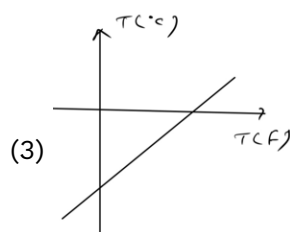
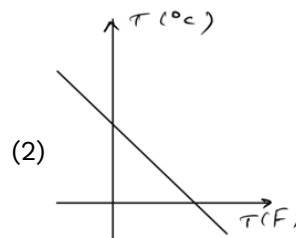
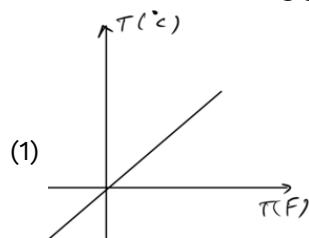
Sol. $\vec{F} = 0$

$\hat{F} \parallel \hat{v}$

$q(\vec{v} \times \vec{B})$

Both (A) & (R) are correct.

Q18 Which of the following graph between $T(^{\circ}\text{C})$ and T in Fahrenheit.



Sol. $\frac{C}{5} = \frac{F - 32}{9}$
 $5F - 160 = 9C$
 $C = \frac{5}{9}F - \frac{160}{9}$
 Ans. (3)

Q19 Arrange UV rays, γ rays, X rays and Infrared rays in ascending order of wavelength.

Sol. γ - rays < X-rays < UV-rays < IR-rays

Q20 An electron is performing circular motion of time period 75 ns inside the solenoid. Current in solenoid is 1.5 A, then find the number of turns per unit length in solenoid.

Sol. $T = \frac{2\pi m}{eB}$
 $T = \frac{2\pi m}{e\mu_0 n I}$

$$n = \frac{2\pi m}{Te\mu_0 l}$$
$$= \frac{2 \times \pi \times 9.1 \times 10^{-31}}{75 \times 10^{-9} \times 1.6 \times 10^{-19} \times 4\pi \times 10^{-7} \times 1.5} = 252$$